



Northern Beaches Secondary College
Manly Selective Campus

2007
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators and templates may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1 - 7
- All questions are of equal value

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Marks

Question 1 Use a SEPARATE writing booklet.

(12)

- a) In how many ways can a committee of three girls and two boys be chosen from eight girls and four boys? 1
- b) Find the Cartesian equation of the curve defined by the parametric equations $x = \sin \theta$ and $y = \cos^2 \theta - 3$. 2
- c) Solve the inequality $\frac{x^2}{x-2} \geq -1$ 3
- d) Point A has coordinates $(7, 4)$ and P has coordinates $(4, -2)$.
Point P divides the interval joining AB externally in the ratio of $3 : 2$.
Determine the coordinates of point B . 3
- e) Use the substitution $t = u^2 - 1$ to evaluate $\int_0^1 \frac{t}{\sqrt{1+t}} dt$ 3

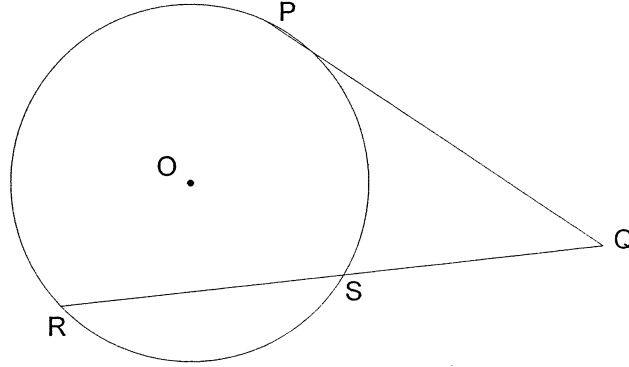
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Question 2 Use a SEPARATE writing booklet.

(12)

- a) In the circle centre O, the tangent PQ is 4 cm. The secant RQ is x cm and the chord RS is y cm.



- i) Show that $y = x - \frac{16}{x}$ 1
- ii) Show that as x increases, so does y . 2
- b) $P(2ap, ap^2)$, $Q(2aq, aq^2)$ and $R(2ar, ar^2)$, where $p < q < r$, are three points on the parabola $x^2 = 4ay$. The tangent to the parabola at point Q has a gradient of q . 3
- If the chord PR is parallel to the tangent at Q , show that p , q and r are consecutive terms in an arithmetic sequence.
- c) Find the exact value of $\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx$ 3
- d) The equation $\ln x = \sin x$ has a first approximation for its solution at $x = 2.5$. Use Newton's method once to find a better approximation. 3

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Question 3 Use a SEPARATE writing booklet.

(12)

a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{3x}$.

1

b) Evaluate $\int_0^{\frac{\pi}{4}} \sin x \cos^2 x \, dx$ and express your answer in the simplest surd form.

3

c) Find $\int \cos^2 9x \, dx$

2

* d) A spherical balloon is expanding so that its volume is increasing at the rate of $24 \text{ cm}^3/\text{s}$. Determine the rate at which its surface area is increasing when the radius is 8 cm.

3

e) A particle is moving with acceleration given by $\ddot{x} = -2\sin x \text{ cm/s}^2$.

3

Initially the particle is at the origin and its velocity, $v = 4 \text{ cm/s}$.

Show that $v = +2\sqrt{\cos x + 3}$

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Question 4 Use a SEPARATE writing booklet.

(12)

- a) i) Use the expansion of the equation

2

$$(1+x)^{n+1} = (1+x)(1+x)^n \text{ to show that :}$$

$$\binom{n+1}{2} = \binom{n}{1} + \binom{n}{2}$$

- (ii) By differentiation of $(1+x)^{2n}$ show that

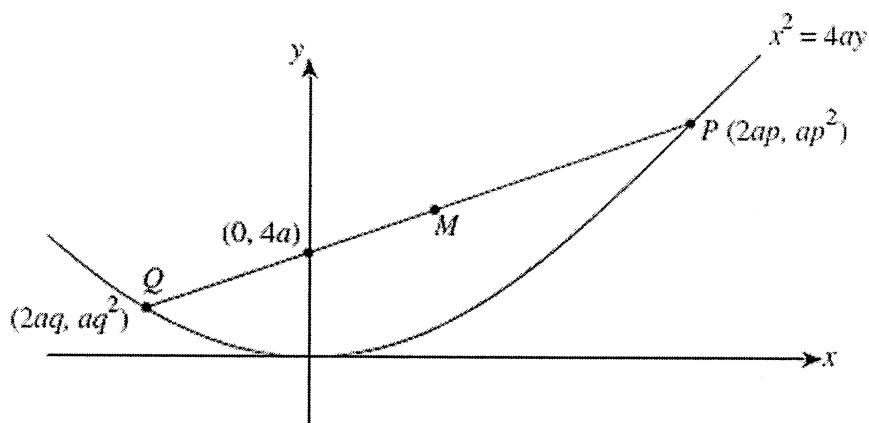
$$\binom{2n}{1} + 2\binom{2n}{2} + 3\binom{2n}{3} + \dots + 2n\binom{2n}{2n} = n \cdot 4^n$$

2

- b) Show that $\tan 2x + \tan x = \frac{\sin 3x}{\cos 2x \cos x}$

2

- c) The diagram shows the graph of the parabola $x^2 = 4ay$.
 The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola.



- (i) Determine the coordinates of M , the midpoint of the chord PQ .

1

- (ii) The chord PQ passes through the point $(0, 4a)$. Show that $pq = -4$

3

- (iii) Hence determine the locus of M as P and Q move along the parabola.

2

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Question 5 Use a SEPARATE writing booklet.

(12)

- a) Use mathematical induction to prove that, for $n \geq 1$ 4

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n-1) \times 2n = \frac{1}{3}n(n+1)(4n-1)$$

- b) Consider the function $y = 2\sin^{-1} 3x$

(i) State its domain and range. 2

(ii) Sketch the function. 1

- (iii) Find the area, in the first quadrant, between the curve, the y -axis
and $y = 0$ and $y = \pi$. 2

- c) A bottle of medicine which is initially at a temperature of 10°C is placed in a room which has a constant temperature of 25°C . The medicine warms at a rate proportional to the difference between the temperature of the room and the temperature (T) of the medicine.
That is, T satisfies the equation

$$\frac{dT}{dt} = -k(T-25) .$$

- (i) Show that $T = 25 + Ae^{-kt}$ satisfies this equation. 1

- (ii) If the temperature after ten minutes is 16°C , find the value of k . 2

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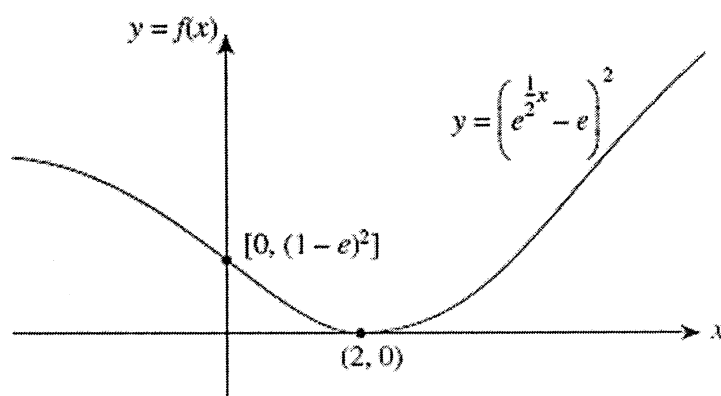
Question 6 Use a SEPARATE writing booklet.

(12)

- a) In Aunt Emily's pantry, the probability that any one item will be past the use-by date is 0.3. Determine the probability that of 12 items in Aunt Emily's pantry, no more than two will be past the use-by date. Express your answer as a decimal correct to two decimal places.

2

- b) The diagram shows the graph of $f(x) = \left(\frac{x}{e^2} - e\right)^2$.



- (i) Explain mathematically why the inverse of

2

$f(x) = \left(\frac{x}{e^2} - e\right)^2$ is not a function.

- (ii) What is the largest domain that includes $x = e$ for which

1

$f(x) = \left(\frac{x}{e^2} - e\right)^2$ has an inverse function, $f^{-1}(x)$?

- (iii) Determine the equation of the inverse function, $f^{-1}(x)$

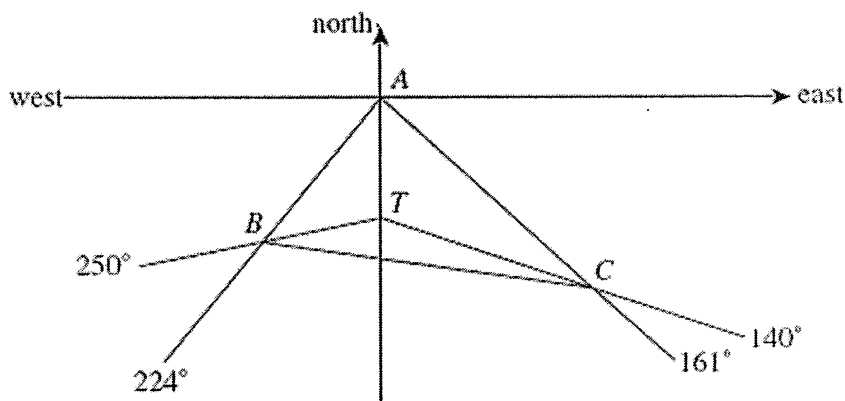
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Marks

Question 6 (*continued*)

- c) From point A , the bearings of B and C are 224° and 161° respectively.
 From point T , 30 km due south of A , the bearings to B and C are 250° and 140° respectively.



- (i) Show that the distance from B to C in kilometres is given by 3

$$(BC)^2 = 900 \left[\left(\frac{\sin 44}{\sin 26} \right)^2 + \left(\frac{\sin 19}{\sin 21} \right)^2 - \frac{2 \sin 44 \sin 19 \cos 110}{\sin 26 \sin 21} \right]$$

- (ii) Hence, or otherwise, determine the time it will take to sail from B to C at an average speed of 10 km/h. 2

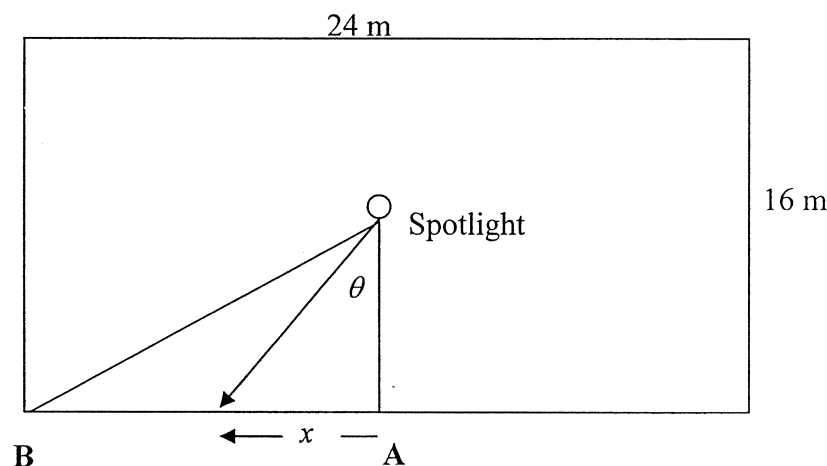
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Question 7 Use a SEPARATE writing booklet.

(12)

- (a) Consider the curve $f(x) = x^3 - 3Bx^2 + 24$ where B is an integer with a constant value.
- (i) Show that the curve has stationary points at $x = 0$ and $x = 2B$.
Determine the nature of the stationary points. 3
- (ii) The polynomial $y = f(x)$ has one root and only one root - at $x = \alpha$.
Show, by graphical means or otherwise, that it is impossible for α to have a positive value. 2
- (iii) Determine the value(s) B can take when α is the only root and $\alpha < 0$. 2
- (b) A spotlight is in the centre of a rectangular nightclub which measures 24 m by 16 m. It is spinning at a rate of 20 rev/min. Its beam throws a spot which moves along the walls as it spins.



- (i) Write the rate of rotation $\frac{d\theta}{dt}$ in radians/sec. 1
- (ii) The spot moves along the wall from A to B at a velocity of $\frac{dx}{dt}$. 2
- Show that $\frac{dx}{dt} = \frac{16\pi}{3} \left(1 + \frac{x^2}{64} \right)$
- (iii) What is the difference in the velocities at which the spot appears to be moving at the points A, nearest to the light and B, furthest from the light? 2

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0 \quad \text{Where } \ln x = \log_e x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Question 1

(a)	${}^8C_3 \times {}^4C_2 = 336$	1 mark: correct answer
(b)	$x = \sin \theta$ <i>i.e.</i> $x^2 = \sin^2 \theta$ $y = \cos^2 \theta - 3$ <i>i.e.</i> $y + 3 = \cos^2 \theta$ <i>but</i> $\sin^2 \theta + \cos^2 \theta = 1$ $\therefore x^2 + y + 3 = 1$ <i>i.e.</i> $y = -x^2 - 2$	2 marks: correct working and answer 1 mark: correct method with an error
(c)	Consider $\frac{x^2}{x-2} = -1$ $x \neq 2$, $x^2 = -x + 2$ $x^2 + x - 2 = 0$ $(x+2)(x-1) = 0$ Critical points are $-2, 1$ and 2 Test $x = 3$, true test $x = 1\frac{1}{2}$, false test $x = 0$, true test $x = -3$, false $\therefore$ solution is $x > 2$ or $-2 \leq x \leq 1$	3 marks: correct soln by a valid method 2 marks: correct method but with an error 1 mark: stating x cannot equal 2
(d)	$A(7,4), P(4,-2), B(x,y)$ $m:n=3:-2$ $(4,-2) = \left(\frac{3x-2.7}{3-2}, \frac{3y-2.4}{3-2}\right)$ $\therefore 3x - 14 = 4$ $3y - 8 = -2$ $3x = 18$ $3y = 6$ $x = 6$ $y = 2$ $\therefore P = (6,2)$	3 marks: correct soln by a valid method 2 marks: correct method with an error 1 mark: correct method with multiple errors, or formula but wrong substitution

(e)	$t = u^2 - 1$ $dt = 2udu$ $t = 1$ $u = \sqrt{2}$ $t = 0$ $u = 1$ $\int_0^1 \frac{tdt}{\sqrt{1+t}} = \int_1^{\sqrt{2}} (u^2 - 1) \frac{2udu}{u}$ $= 2 \int_1^{\sqrt{2}} (u^2 - 1) du$ $= 2 \left[\frac{u^3}{3} - u \right]_1^{\sqrt{2}}$ $= 2 \left[\left(\frac{2\sqrt{2}}{3} - \sqrt{2} \right) - \left(\frac{1}{3} - 1 \right) \right]$ $= \frac{-2\sqrt{2}}{3}$	3 marks: correctly substituting and finding and evaluating integral 2 marks: correct method, mistake in evaluation, or incorrect limits 1 mark: showing correct substitution
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Question 2

(a) (i)	$QP^2 = QS \times QR$ $4^2 = (x-y) \times x$ $\frac{16}{x} = x-y$ $y = x - \frac{16}{x}$	1 mark – correct proof
(ii)	EITHER $\frac{dy}{dx} = 1 + \frac{16}{x^2}$ $\therefore \text{ as } x \rightarrow \infty \frac{dy}{dx} \rightarrow \infty$ Hence y is an increasing function in x OR $y = x - \frac{16}{x}$ $\text{as } x \rightarrow \infty \frac{16}{x} \rightarrow 0$ $\therefore y \rightarrow x \text{ and as } x \rightarrow \infty y \rightarrow \infty$	2 marks – correct demonstration 1 mark – demonstration approached correctly
(b)	Gradient of PR = $\frac{ap^2 - ar^2}{2ap - 2ar}$ $= \frac{p+r}{2}$ Gradient at Q is q (by definition) $\therefore \frac{p+r}{2} = q$ – so q is the arithmetic mean of p and r OR let $q = p + d$ and $r = p + 2d$ $\therefore \frac{p+r}{2} = \frac{p+p+2d}{2}$ $= p + d = q$ $\therefore p, q \text{ and } r \text{ are in arithmetic progression}$	3 marks – correct demonstration 2 marks – gradients equated but subsequent error 1 mark – gradients established correctly

Question 2 (continued)

(c)	$\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx = \int_0^1 \frac{1}{\sqrt{3} \times \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 - x^2}}$ $= \frac{1}{\sqrt{3}} \left[\sin^{-1} \left(\frac{\sqrt{3}x}{2} \right) \right]_0^1$ $= \frac{1}{\sqrt{3}} \left(\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1}(0) \right)$ $= \frac{1}{\sqrt{3}} \times \frac{\pi}{3} = \frac{\pi}{3\sqrt{3}}$	3 marks – correct answer 2 marks – correct integration but substitution error OR 1/a coefficient used by mistake and substitution correct 1 mark – fraction transformed appropriately
(d)	Equation is $f(x) = \ln x - \sin x$ $f'(x) = \frac{1}{x} - \cos x$ Remember as $x = 2.5$, this measure is in radians $f(2.5) = \ln 2.5 - \sin 2.5$ (radians) $= 0.3178$ $f'(2.5) = 1.2011$ $x_2 = 2.5 - \frac{0.3178}{1.2011} = 2.235$ NOT 3.9568 if x taken as degrees	3 marks – correct answer 2 marks – correct differentiation and substitution but 2.5 used as degrees 1 mark – correct differentiation but subsequent error
	Comments: (a ii) and (b) Marks were given only when statements were precise. Too often assumptions were substituted into a proof and even actual numbers were substituted to establish a trend – against all what has been said. (c) The coefficient for the square root must be explicitly considered by many as most students had no idea of what it was – or even if it existed!!!	

Question 3

(a)	$\frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{x}$ $= \frac{4}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}$ $= \frac{4}{3} \times 1$ $= \frac{4}{3}$	1 mark - Gives the correct answer, with no wrong method displayed
(b)	$\int_0^{\frac{\pi}{4}} \sin x \cos^2 x \, dx = -\left[\frac{1}{3} \cos^3 x\right]_0^{\frac{\pi}{4}}$ $= -\frac{1}{3} \left[\cos^3 \frac{\pi}{4} - \cos^3 0\right]$ $= \frac{2\sqrt{2}-1}{6\sqrt{2}} \text{ or } \frac{4-\sqrt{2}}{12}$	2 marks - Gives the correct answer 1 mark - Gives the correct integral
(c)	$\int \cos^2 9x \, dx = \int \frac{1}{2} (1 + \cos 18x) \, dx$ $= \frac{1}{2} \left(x + \frac{1}{18} \sin 18x \right) + c$ $= \frac{x}{2} + \frac{1}{36} \sin 18x + c$	2 marks – correct answer 1 mark – correct change of $\cos^2 9x$ to $\frac{1}{2}(1 + \cos 18x)$
(d)	$\frac{dV}{dt} = 24 \text{ cm}^3/\text{s}, r = 8$ $\frac{dSA}{dt} = \frac{dSA}{dr} \times \frac{dr}{dv} \times \frac{dv}{dt}$ $= 8\pi r \times \frac{1}{4\pi r^2} \times 24$ $= \frac{48}{r}$ $= 6 \text{ cm}^2/\text{s}$	3 marks - Gives the correct answer 2 marks - Determines $\frac{dSA}{dt} = \frac{dSA}{dr} \times \frac{dr}{dv} \times \frac{dv}{dt}$ or equivalent. Uses one piece of information correctly. 1

Question 3 (continued)

(e)	$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -2 \sin x$ $\frac{1}{2} v^2 = 2 \cos x + C$ given $t = 0, x = 0, v = 4$ $8 = 2 \cos 0 + C$ $C = 6$ $\frac{1}{2} v^2 = 2 \cos x + 6$ $v^2 = 4 \cos x + 12$ $v = \pm 2 \sqrt{\cos x + 3}$ but $v > 0$ when $x = 0$, so + is required $\therefore v = + 2 \sqrt{\cos x + 3}$	3 marks - Gives a correct demonstration 2 marks - Gives an essentially correct demonstration that omits checking $\pm$ or similar 2 • Uses the ... $\frac{d\left(\frac{1}{2} v^2\right)}{dx}$ 1
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Question 4

(a) (i)	$(1+x)^{n+1} = \binom{n+1}{0}x^0 + \binom{n+1}{1}x^1 + \binom{n+1}{2}x^2 \dots$ $(1+x)(1+x)^n = (1+x)\left(\binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \dots\right)$ Equating coefficients of x^2 $\binom{n+1}{2} = \binom{n}{2} + 1 \times \binom{n}{1}$	2 marks – correct proof 1 mark – correct expansions
(ii)	$(1+x)^{2n} = \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \binom{2n}{3}x^3 + \dots + \binom{2n}{2n}x^{2n}$ Differentiating both sides $2n(1+x)^{2n-1} = \binom{2n}{1} + 2\binom{2n}{2}x + 3\binom{2n}{3}x^2 + \dots + 2n\binom{2n}{2n}x^{2n-1}$ Let $x=1$ $2n \times 2^{2n-1} = \binom{2n}{1} + 2\binom{2n}{2} + 3\binom{2n}{3} + \dots + 2n\binom{2n}{2n}$ But $2n \times 2^{2n-1} = n \times 2^{2n} = n \times 4^n$ $\therefore n \times 4^n = \binom{2n}{1} + 2\binom{2n}{2} + 3\binom{2n}{3} + \dots + 2n\binom{2n}{2n}$	2 marks – correct proof 1 mark – correct expansions
(b)	$\text{RHS} = \frac{\sin 3x}{\cos 2x \cos x}$ $= \frac{\sin 2x \cos x + \cos 2x \sin x}{\cos 2x \cos x}$ $= \tan 2x + \tan x = \text{LHS}$	2 marks – correct proof 1 mark – correct expansions

Question 4 (continued)

(c)(i)	Midpoint is $X = \frac{2ap + 2aq}{2} = a(p+q)$ $Y = \frac{ap^2 + aq^2}{2} = \frac{a(p^2 + q^2)}{2}$	1 mark – both coordinates correct
(ii)	Eqn of PQ is $\frac{y - ap^2}{aq^2 - ap^2} = \frac{x - 2ap}{2aq - 2ap}$ If passing through (0,4a) $\frac{a(4 - p^2)}{a(q - p)(q + p)} = \frac{-2ap}{2a(q - p)}$ $4 - p^2 = -(pq + p^2)$ $pq = -4$	3 marks – correct proof 2 marks – substitution correct but subsequent error 1 mark – equation formed
(iii)	$p + q = \frac{X}{a}$ $\frac{2Y}{a} = p^2 + q^2 = (p+q)^2 - 2pq$ $= \frac{X^2}{a^2} - 2(-4)$ $2aY = X^2 + 8a^2$	2 marks – correct solution 1 mark – combined equation formed

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2007 HSC Mathematics Extension 1 – Trial - solutions

Question 5

(a)	Prove that $1 \times 2 + 3 \times 4 + \dots + (2n-1) \times 2n = \frac{1}{3}n(n+1)(4n-1)$ Step 1: Let $n = 1$ LHS = 1×2 RHS = $\frac{1}{3}1 \times 2 \times 3$ $\therefore$ true for $n = 1$ Step 2: Assume true for $n=k$ i.e. $1 \times 2 + 3 \times 4 + \dots + (2k-1) \times 2k = \frac{1}{3}k(k+1)(4k-1)$ Try to show true for $n=k+1$, i.e. that $1 \times 2 + 3 \times 4 + \dots + (2k-1) \times 2k + (2k+1)(2k+2) =$ $\frac{1}{3}(k+1)(k+2)(4k+3)$ LHS = $1 \times 2 + 3 \times 4 + \dots + (2k-1) \times 2k + (2k+1)(2k+2)$ $= \frac{1}{3}k(k+1)(4k-1) + (2k+1)(2k+2)$ from assumption $= (k+1) \left[\frac{1}{3}k(4k-1) + 2(2k+1) \right]$ $= \frac{k+1}{3} [4k^2 - k + 12k + 6]$ $= \frac{k+1}{3} [4k^2 + 11k + 6]$ $= \frac{k+1}{3} (k+2)(4k+3)$ $= \text{RHS} \therefore$ if true for $n=k$ then true for $n=k+1$ Step 3: But true for $n=1$, $\therefore$ true for $n=1+1=2$, and similarly for $n=3, 4, 5, \dots$ and all positive integral n	4 marks: correct soln 3 marks: correct method but with one error in algebra 2 marks: $n=1$ correct and correct $n=k+1$ line, multiple errors in algebra 1 mark: correctly showing for $n=1$
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(b)(i)	$y = 2 \sin^{-1} 3x$ for domain $-1 \leq 3x \leq 1$ i.e. $-\frac{1}{3} \leq x \leq \frac{1}{3}$ for range $-\frac{\pi}{2} \leq \sin^{-1} 3x \leq \frac{\pi}{2}$ i.e. $-\pi \leq y \leq \pi$	1 mark: correct domain 1 mark: correct range 1 mark: correct diagram
(iii)	Area = $\int_0^{\pi} x \, dy$ $y = 2 \sin^{-1} 3x$ $x = \frac{1}{3} \sin\left(\frac{y}{2}\right)$ $= \int_0^{\pi} \frac{1}{3} \sin\left(\frac{y}{2}\right) dy$ $= \left[-\frac{1}{3} \frac{1}{\frac{1}{2}} \cos\left(\frac{y}{2}\right) \right]_0^{\pi}$ $= \left(-\frac{2}{3} \cos\left(\frac{\pi}{2}\right) \right) - \left(-\frac{2}{3} \cos 0 \right)$ $= 0 - \left(-\frac{2}{3} \right)$ $= \frac{2}{3}$	2 marks: correct answer by appropriate working 1 mark: correct method but with one mistake
(c)(i)	$T = 25 + A e^{-kt}$ $\frac{dT}{dt} = -k A e^{-kt}$ but $A e^{-kt} = T - 25$ $\therefore \frac{dT}{dt} = -k(T - 25)$ as required	1 mark: correctly showing
(ii)	when $t=0$, $T=10$, $\therefore 10 = 25 + A e^0$ i.e. $A = -15$ when $t=10$, $T=16$, $\therefore 16 = 25 - 15 e^{-10k}$ $15 e^{-10k} = 9$ $e^{-10k} = \frac{3}{5}$ $-10k = \ln\left(\frac{3}{5}\right)$	2 marks: correctly finding value of k 1 mark: correct value of A

Question 6

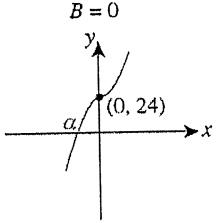
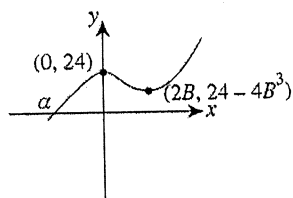
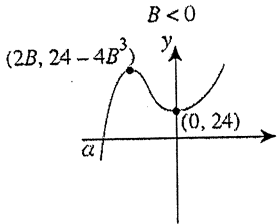
(a)	$\Pr(\text{use-by}) = 0.3$ $\Pr(\text{no more than 2}) = \Pr(0 \text{ items}) + \Pr(1 \text{ item}) + \Pr(2 \text{ items})$ $= {}^{12}C_0 (0.3)^0 (0.7)^{12} + {}^{12}C_1 (0.3)^1 (0.7)^{11} + {}^{12}C_2 (0.3)^2 (0.7)^{10}$ $= 0.25$	2 marks - correct answer 1 mark - correct statement with binomial probability but arithmetic error
(b)	(i) The inverse of $f(x)$ is not a function because there is a turning point in the domain. Hence for any given value on the range, there are two possible x values. An inverse of $f(x)$ ($f^{-1}(x)$) therefore has two possible y values for a given x value which does not conform to the definition of a function. (ii) For the function to be an inverse, the function on either side of the stationary point must be defined. Hence the largest domain is $x \geq 2$ or $x \leq 2$ NOTE: the answer $x > 2$ is not acceptable.	2 marks - complete answer 1 mark - partially correct statement 1 mark - correct domain
(iii)	$y = \left(e^{\frac{1}{2}x} - e\right)^2$ $x = \left(e^{\frac{1}{2}y} - e\right)^2$ $\sqrt{x} = e^{\frac{1}{2}y} - e$ $e^{\frac{1}{2}y} = e + \sqrt{x}$ $y = 2\ln(e + \sqrt{x})$	2 marks - inverse correctly determined 1 mark - interchange of x and y and some progress made to determine inverse.
	NOTES: In part (i), a common error resulted from misreading the question – the emphasis should have been on the INVERSE not being a function but many answers test the original equation and showed that was (or was not) a function. In addition, there is no such thing as the “horizontal line test” and so it cannot be quoted as evidence.	

Question 6 (continued)

(c)	In these questions, always begin by analysing the diagram to identify where the angles are which are listed in the “to prove” statement.	3 marks - correct proof for BC^2 2 marks – expressions determined for BT and CT but subsequent error or lack of justification. 1 mark – correct statement for either BT or TC or significant attempt (with justification) in either triangle to develop the sine rule relationship.
(i)	First use the sine rule to $\triangle ABT$ to find BT $BT = 30 \times \frac{\sin 44^\circ}{\sin 26^\circ}$ Then examine $\triangle ATC$ to find TC $TC = 30 \times \frac{\sin 19^\circ}{\sin 21^\circ}$ Now using $\triangle BTC$ with the cos rule focussing on $\angle BTC = 70^\circ + 40^\circ = 110^\circ$ $BC^2 = BT^2 + TC^2 - 2 \times BT \times TC \times \cos BTC$ $= \left(30^2 \times \frac{\sin^2 44^\circ}{\sin^2 26^\circ}\right) + 30^2 \times \left(\frac{\sin 19^\circ}{\sin 21^\circ}\right)^2$ $- 2 \times \left(\frac{\sin 44^\circ}{\sin 26^\circ}\right) \times \left(\frac{\sin 19^\circ}{\sin 21^\circ}\right) \times \cos 110^\circ$	
(ii)	$BC = 62.36$ Hence at 10km/hr time = 6.236 hours = 6 hours 14 minutes NOTE: In (c) (i) it is essential that all angles used are shown and proved to have the values attributed. This can be done even through a diagram.	2 marks - correct answer 1 mark - correct answer for BC reported

Question 7

7 (a) - i	(i) $f'(x) = 3x^2 - 6Bx = 0$ $3x(x - 2B) = 0$ $\therefore x = 0$ or $x = 2B$ Therefore, the stationary points are $(0, 24)$ and $(2B, 24 - 4B^3)$ $f''(x) = 6x - 6B$ When $B > 0$ $f''(0) < 0$ $(0, 24)$ maximum $f''(2B) > 0$ $(2B, 24 - 4B^3)$ minimum When $B < 0$ $f''(0) > 0$ $(0, 24)$ minimum $f''(2B) < 0$ $(2B, 24 - 4B^3)$ maximum When $B = 0$ $(0, 24)$ will be a horizontal inflexion	3 marks – full answer. 2 marks – for considering B either neg. or pos. 1 mark – showing stationary points but not taking into account different possibilities for B
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7 (a) - ii	There are three possible graphs for $y = f(x)$ with exactly one root. $B = 0$  $B > 0$  $B < 0$  In each case, $\alpha < 0$. Therefore, α cannot be positive.	2 marks – complete answer 1 mark – explaining one case only of three possible cases.
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7 (a) - iii	When $B = 0$, there is a valid solution (see graph). When $B > 0$, $0 < 24 - 4B^3 < 24$ $-24 < -4B^3 < 0$ $\therefore 0 < 4B^3 < 24$ $0 < B^3 < 6$ $0 < B < \sqrt[3]{6}$ As B is an integer, B could be 1. When $B < 0$, obviously from the graph all values will satisfy. algebraically, $24 - 4B^3 > 24$ $\therefore -4B^3 > 0$ $B < 0$ Thus, as B is an integer, B could have the values $\{1, 0, -1, -2 \dots\}$.	2 marks – correct answer. 1 mark – attempt to use $24 - 4B^3$ in a correct manner.
(b) - i	$20 \text{ rev/min} = 20 \times 2\pi \text{ rad/min}$ $= \frac{40\pi}{60} = \frac{2\pi}{3} \text{ rad/sec}$	1 mark – correct answer
(b) - ii	$\tan \theta = \frac{x}{8}$ $x = 8 \tan \theta$ $\frac{dx}{d\theta} = 8 \sec^2 \theta$ $\frac{dx}{dt} = \frac{dx}{d\theta} \cdot \frac{d\theta}{dt}$ $= 8 \sec^2 \theta \cdot \frac{2\pi}{3}$ $= \frac{16\pi}{3} \sec^2 \theta$ $= \frac{16\pi}{3} (1 + \tan^2 \theta)$ $= \frac{16\pi}{3} (1 + \tan^2 \theta)$ $= \frac{16\pi}{3} \left(1 + \left(\frac{x}{8} \right)^2 \right)$ $= \frac{16\pi}{3} \left(1 + \frac{x^2}{64} \right)$	2 marks for full solution 1 mark if done in terms of θ or otherwise incomplete

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(b) - iii	At B, $x = 12$ $\frac{dx}{dt} = \frac{16\pi}{3} \left(1 + \left(\frac{12}{8} \right)^2 \right)$ $= \frac{16\pi}{3} \left(\frac{13}{4} \right)$ $= \frac{52\pi}{3}$ At A, $x = 0$ $\frac{dx}{dt} = \frac{16\pi}{3} \left(1 + \left(\frac{0}{8} \right)^2 \right)$ $= \frac{16\pi}{3}$ Difference = $\frac{52\pi - 16\pi}{3}$ $= 12\pi \text{ m/s}$	2 marks – correct answer 1 mark if only one found correctly or if subtraction incorrect.
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